Supporting teleworking with multimedia

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The term 'teleworking' covers a variety of ways of working flexibly — at home, on the move, even in the office. Just like other workers, teleworkers need a support infrastructure to enable them to work efficiently. Multimedia applications have proved to be an ideal support tool for teleworking. This paper examines BT's experience of using multimedia in a teleworking context.

1. Introduction

Teleworking is a flexible way of working which covers a wide range of work activities, all of which entail working remotely from an employer, or normally expected place of work, for a significant proportion of work time. Teleworking may be on either a full-time or a part-time basis. The work often involves the electronic processing of information, but always involves using telecommunications to keep in contact with the remote employer.

This definition excludes the traditional 'outworkers', as well as people who work at home for a day now and then, but includes a wide variety of working situations:

- people working at home (e.g. programmers),
- people working from home (e.g. salespeople),
- people working at work centres (such as telecottages, or satellite offices).

Major companies are showing increasing interest in the use of teleworking because of the benefits it can bring to the company and its employees. The company benefits include some (but not usually all) of the following:

- better customer service through increased flexibility in the workforce,
- increased productivity, often due to less sick-leave, less interruptions and lack of background office noise,
- skills retention and improved recruitment opportunities,
- reduced overheads usually through savings in office accommodation,
- resilience to power cuts, bomb threats and public transport problems.

However, in order to get the best from a teleworking programme, careful consideration needs to be given at the planning stage to a wide range of issues.

- Management issues
  - — selection criteria,
  - — training,
  - — security,
  - — supervision,
  - — trade unions,
  - — terms and conditions,
  - — support at home.
- Legal considerations
  - — health and safety,
  - — planning regulations,
  - — capital gains tax,

— income tax,

— insurance,

— mortgage and tenancy agreements,

— local council taxes.

- Human aspects

— home environment,

— social isolation,

— child care,

— career advancement.

Technology will have little or no impact on many of these issues, but in other, important, cases it can be used to enhance benefits or overcome difficulties. A key area is the last of the 'management issues' listed above — 'support at home'. It is not enough merely to provide a homeworker with the capability to do their normal work at home. In recognition of the remoteness of the worker, adequate support must be given. Support is the key to successful teleworking. In order to establish how much support is 'adequate', BT Laboratories has been involved in a number of teleworking projects. Of these, the one most directed at investigating the support issue, and the most well-known, is 'The Inverness Experiment' [1].

2. BT's Inverness experiment

The purpose of carrying out teleworking experiments is to explore, in detail, particular working situations and to try out ideas for providing teleworking support using 'leading edge' technology.

As with any experiment, it was important not to change too many variables at once. For this reason, and to allow direct comparison, the work patterns of the operators working at home mirrored those of the Directory Assistance Centre-based operators. In a pilot or trial this constraint would not exist and increased efficiency and flexibility could be achieved through adjusting the working hours of the teleworkers.

The objectives of this teleworking experiment were:

- to demonstrate that call centre operators can successfully work at home,
- to explore how the facilities and support provided for office-based workers can be extended to homeworkers,
- to investigate how the technical and non-technical problems of teleworking can be overcome,
- to look at how support for 'human factors' and supervisory issues can be provided for teleworkers using technology.

2.1 The teleworking system

The experimental teleworking system for the Directory Assistance (DA) operators was developed at BT Laboratories. It was designed so that the job of the teleworking operators remained essentially the same as that of the Centre-based operators. The experiment involved ten BT operators in the Inverness area, where it was possible to take advantage of an extensive ISDN network.

The system was also designed with several types of user in mind and these were the main participants in the teleworking experiment:

- the teleworkers themselves,
- supervisors within the DA Centre,
- senior operators in the DA Centre with responsibility for various miscellaneous clerical and administrative functions,
- Centre-based operators.

Each of these groups interacted with the system via a terminal with each terminal being tailored, in terms of functionality, to meet the requirements of the group it served.

The teleworking system incorporated several support facilities for the teleworkers, based on an extensive study of the needs of the operators. These included:

- videophone — for dealings with supervisors, and for socialising with colleagues when off-duty,
- e-mail — allowing messages to be sent between teleworkers and between the DA Centre and teleworkers,
- electronic noticeboards — for shared information,
- electronic forms — to replace the paper forms used by operators in the DA Centre,
- newsflash — to 'broadcast' urgent information to the homeworkers,
- SOS — for notifying the supervisor of a domestic emergency,

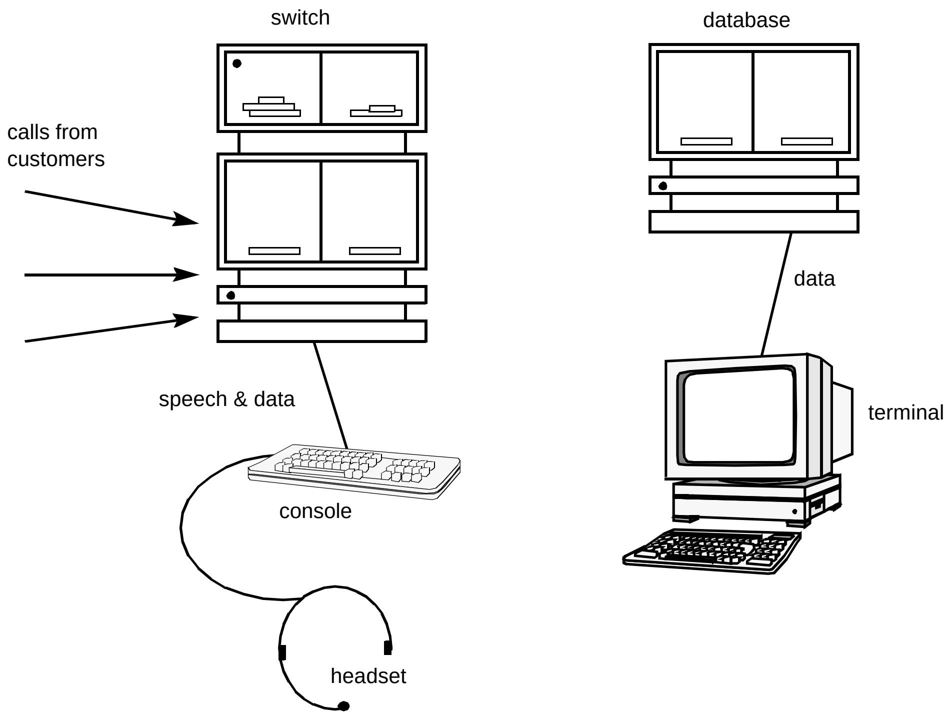

Fig 1 Normal DA Centre-based configuration.

- comfort break — for requesting a comfort break during a shift.

Figure 1 shows the normal configuration of equipment in the Inverness Directory Assistance (DA) Centre as used by the operators based in the DA Centre (see Fig 2).

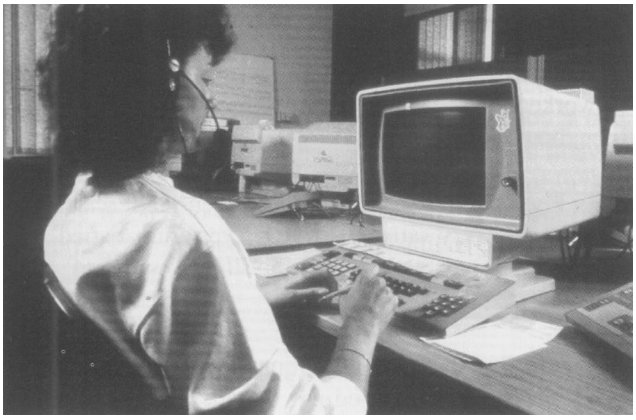

Fig 2 An operator in the DA Centre.

Incoming calls from the customers were distributed to the operators by the automatic call distribution (ACD) switch which gave an even loading amongst the operators and equalised the call waiting time. The operator answered the call from a console, which consisted of a desk-top unit (also known as a 'turret') and headset, and then used a special terminal to interrogate the directory database and find the telephone number required by the customer. All of the components shown in the diagram, apart from the directory database, were located in the DA Centre.

These essential elements, which allowed the operators to do their work, were extended into their homes using ISDN as shown in Fig 3.

However, from the initial feasibility study it was concluded that additional support was essential for remote workers. When the additional support facilities for the experiment were added, the situation became more complicated. Figure 4 illustrates the configuration of equipment which was used to support the teleworking operators.

The support facilities were provided largely by the central support services computer. Multiplexers were used to combine the data and voice channels for transmission across one of the two 64-kbit/s ISDN channels. The videophone, which appeared in a window on the teleworker's screen, used the second channel. The teleworker's terminal was a personal computer which emulated the 'centre' terminal (as shown in Fig 1). The computers all used a UNIX operating system. All the user interfaces used X Windows Motif.

There was a supervisor's terminal, located in the Centre, from where the supervisor could manage and interact with the teleworkers. A second supervisory terminal was provided for the use of the Senior Operator responsible for certain clerical tasks within the DA Centre.

There was also a terminal, with limited functionality, provided in the Centre's welfare room — where the Centre operators usually sit during breaks from work — to allow socialising between the teleworkers and Centre operators.

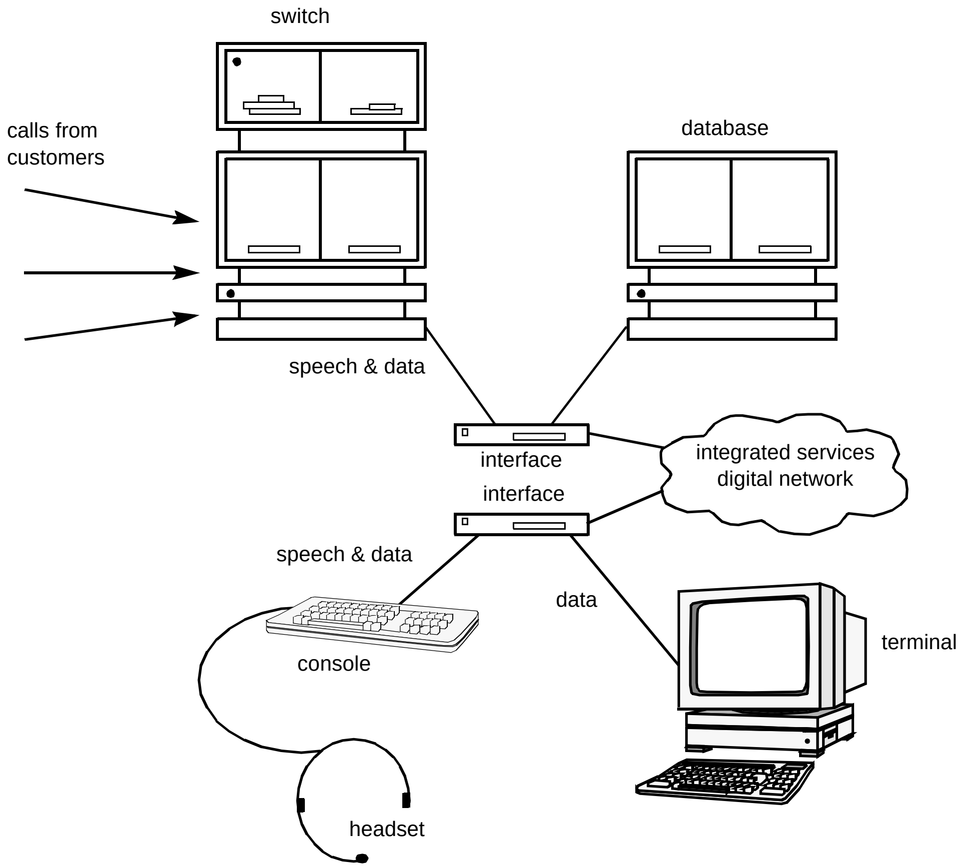

Fig 3 Remote working.

2.2 General considerations

Teleworkers and supervisors were each given 2—3 days training on the use of the teleworking system, prior to the beginning of the experiment in June 1992. Throughout the 12-month experiment the teleworking system was monitored with data being gathered on the usage and perceptions of the support facilities. In addition, the Department of Psychology at the University of Aberdeen tracked the psychological effects on behalf of BT Laboratories.

3. Experiment results

3.1 Teleworkers

The operators who took part in the experiment benefited from savings in time and money through avoiding commuting, reduced stress and greater flexibility. Drawbacks were relatively minor, with the biggest problem being the waiting time for faults to be fixed (see also section 3.4).

Even with these minor drawbacks Fig 5 shows that the teleworkers were generally well satisfied with teleworking throughout the experiment.

In Fig 5 the Y-axis represents a seven-point scale where 7 represents ‘very satisfied’ and 1 ‘ very dissatisfied’. The majority of scores are between 5 ‘slightly satisfied’ and 6 ‘satisfied’.

Figure 5 shows one exceptional score (4 on the chart represents ‘neither satisfied nor dissatisfied’) around week twenty, which was the Christmas period. This was mainly due to resourcing difficulties in the DA Centre over the Christmas period, leading to teleworkers being told not to use the videophone to make contact with the DA Centre before ‘logging-on’ at the start of a shift. Instead they were told to contact a Senior Operator using the telephone and, as a result, did not receive their daily team talks from the supervisor. The teleworkers missed their daily ‘face-to-face meeting’ with their supervisor (see Fig 6).

For the majority of teleworkers, the expected problem of isolation did not materialise. In part, this is due to the range of communications facilities that they had at their disposal,

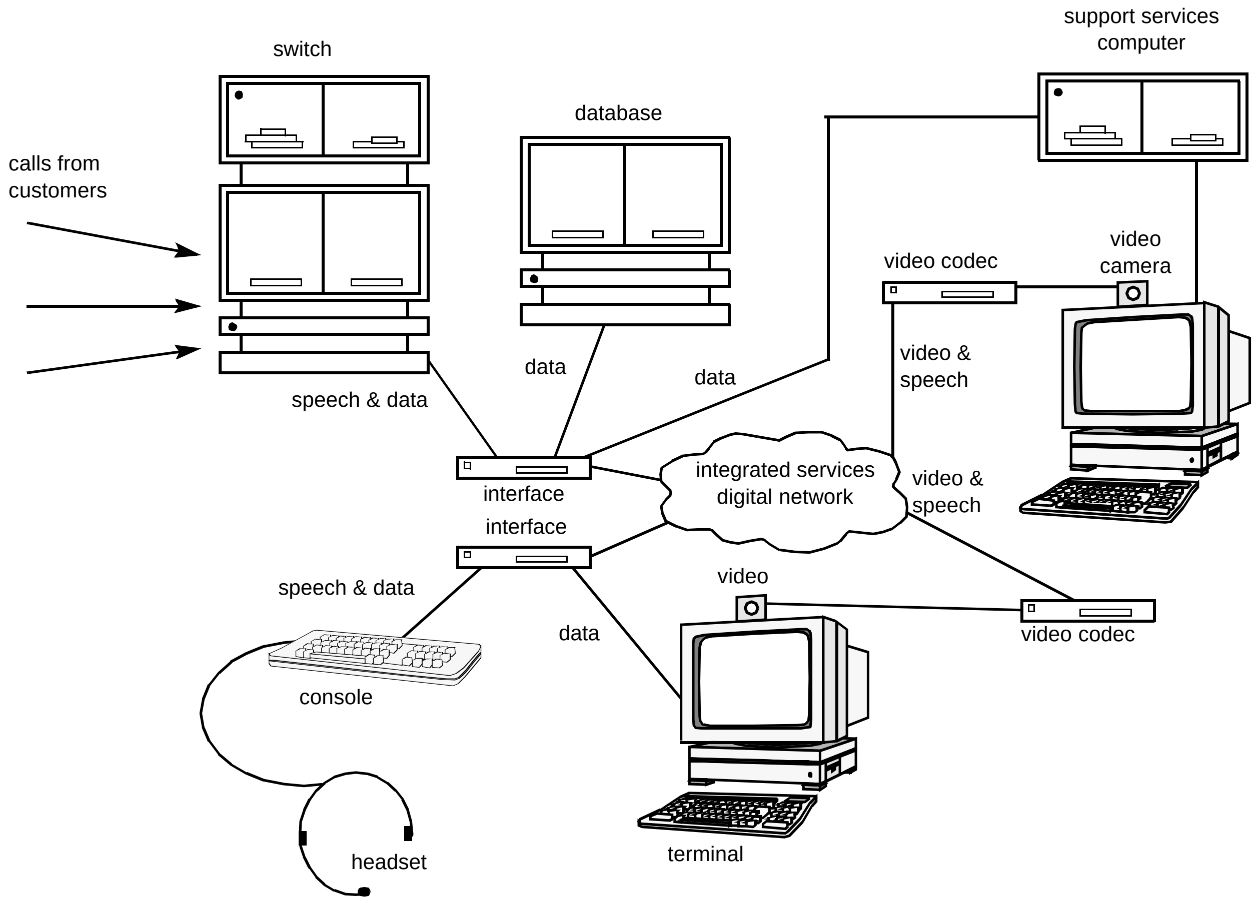

Fig 4 Remote working with support.

but their experience also seems to back up findings, reported from other teleworking programmes, that isolation is largely a perceived problem. For teleworkers who are part of a well-planned teleworking programme, isolation is not a big problem. At the end of the twelve months the majority of teleworkers had found teleworking so beneficial to them that they wanted to continue teleworking. Nine out of ten teleworkers also indicated that their families preferred them teleworking.

3.2 *Company benefits*

The experiment has shown how teleworking can positively affect the service offered to customers and the productivity of operators:

- flexibility — teleworkers were more willing to work outside their standard hours,

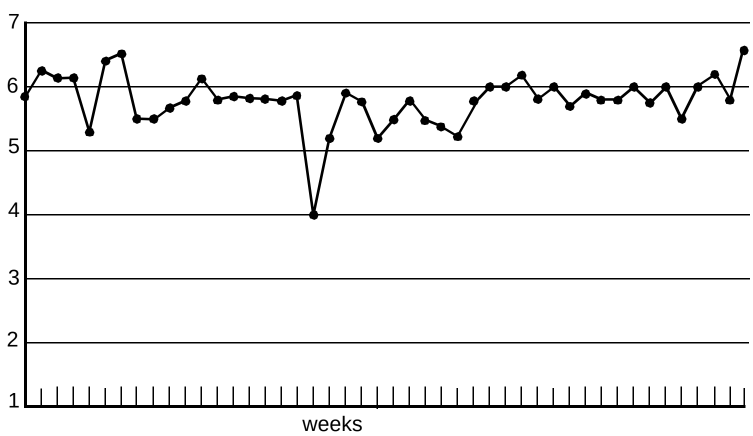

Fig 5 Response to the question: ‘Overall how satisfied have you been with teleworking this week?’

Fig 6 One of the teleworkers at work.

- resilience — teleworkers were unaffected by weather and travel problems,
- skills retention — one operator moved 150 miles away, but continued as part of the experiment,
- productivity — statistics showed that teleworkers were just as productive when working, and they also tended to take less time off through minor illness.

3.3 Technology

Videotelephony was an extremely valuable part of the teleworking system. The videophone was used the most out of all the support facilities and was considered, by the teleworkers, to be the most important, easy to use, enjoyable and easy to learn. It was also the major form of communication that the supervisors used to contact the teleworkers. Some of the operators that worked at home suggested that the videophone would have been used even more if it had been accessible while the rest of the teleworking system was switched off.

A videophone was provided in the welfare room at the DA Centre but it was used very little. This was due to a variety of reasons, including the difficulty of the teleworkers knowing who would be in the welfare room at any given time. However, the main reason seems to have been the reluctance of the Centre-based operators to use the videophone terminal. This was mainly caused by the operators feeling inhibited, possibly through lack of confidence, about using the terminal in front of their colleagues.

E-mail was also seen as important and useful even though the ‘off-the-shelf’ package used in the experiment had a poorer user interface than the other facilities, and was therefore not easy to use or learn. After a fairly slow take-up by the teleworkers, and once the system had been modified to make it easier to access and use, e-mail became an effective and essential means of communication for the teleworkers.

The videophone and e-mail were two of the two-way communications facilities used in the experiment, the others being telephone and post. Figure 7 shows the average number of times that these facilities were initiated by the individual teleworkers and from the terminals used by the supervisors and senior operators.

Of the other support facilities the electronic form and electronic noticeboard facilities were thought to be the next most important after videophone and e-mail.

Overall the experiment system was ranked highly by the teleworkers. When interviewed at the end of the experiment, the teleworkers indicated that there was little of any significance that they would change on or add to the system in order to improve it.

3.4 Lessons learnt

Maintenance

The importance of good maintenance procedures was highlighted during the experiment. In the event of a fault, teleworking operators do not have the option, that Centre-based operators may have, of moving to a spare terminal. Instead the fault must be repaired as quickly as possible, or a decision taken for the operator to travel to the DA Centre. To enable swift repair of the fault the procedures should be such that the fault is quickly diagnosed and focused action taken to repair it.

For the experiment, a good maintenance programme was arranged, with a guaranteed call-out time for service engineers of not more than 4 hours (comparable with BT’s ‘Total Care’ maintenance package). However, for part-timers, 4 hours is a complete shift. Any employer introducing teleworking for people whose work is dependent on the availability of ‘information technology’ must consider the balance between the cost of fast

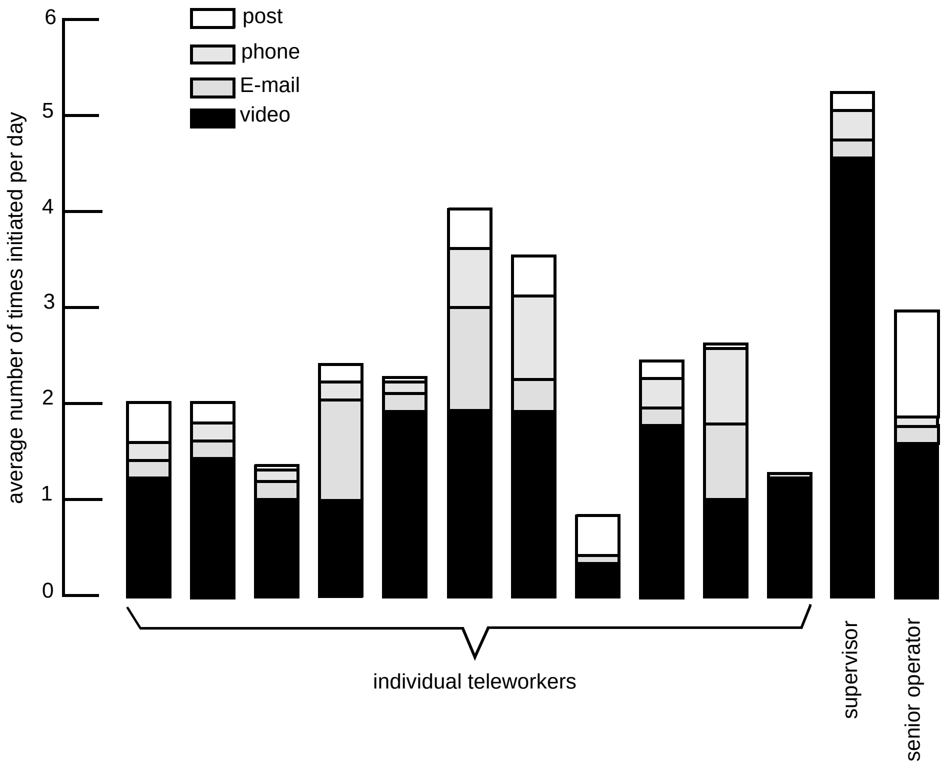

Fig 7 Average usage of the two-way communications facilities offered by the teleworking system.

maintenance response and the wasted time of an employee unable to work.

Supervision

The supervisor/teleworker communication channel is an important one. It is important both in terms of supervision and because the supervisor is the main point of contact for the teleworkers. During the experiment supervisors needed to spend more time giving daily team talks to the teleworkers. Team talks were given using the videophone and therefore had to be repeated for each teleworker. Multipoint videotelephony would overcome this problem.

Socialising

For the majority of the teleworkers, the expected problem of isolation did not materialise. In part this is due to the range of communications facilities that they had at their disposal, but their experience also seems to back up findings, reported from other teleworking programmes, that isolation is largely a perceived problem. Potential teleworkers expect isolation to be a problem, but find in practice that it is not. This is exemplified in the experiment by the fact that the teleworkers preferred to use free time, such as regular breaks from work, to 'do their own thing' rather than socialise over the network.

4. Expanding the teleworking arena

The Inverness experiment showed that teleworking was possible, and beneficial, in a call centre environment. BT has followed up on the success of this experiment by introducing teleworking in its own call centres. Earlier this year, one sales area started the ball rolling with 12 people on the '152' service based around the Southampton area. The principles of support, and of the technology used, follow closely the Inverness pattern. The major difference is that all the equipment and software is now available 'off-the-shelf'.

The experiment stimulated interest in the possibilities among a number of companies who have 'telephone enquiry agents' working in call centres. BT is now able to provide consultancy on all teleworking issues to help such companies introduce a teleworking programme.

However, the principles of support developed for Inverness can be applied to a much wider range of jobs, although the technology used to give that support may vary. For example, for professional teleworkers, such as the people at BT Laboratories, or colleagues in marketing and sales, the screen-sharing facilities of BT's VC8000 Multimedia Communications Card could be invaluable. Other 'groupware' products, like Microsoft 'Windows for Workgroups' and Lotus 'Notes', will help distributed teams work as one unit.

The time has arrived for teleworking. Companies need its benefits for competitive advantage; individuals desire the freedom and flexibility it can give them. The technology to support teleworking is well established, and getting more extensive by the day. Only old-fashioned management attitudes hinder the growth of teleworking.

Acknowledgements

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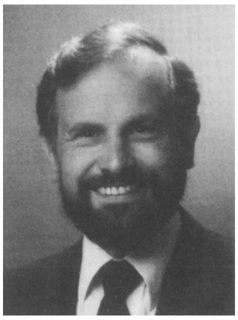
Reference

1 BT Report: 'The Inverness experiment', available from BT Laboratories (Tel: (01473) 646044, Fax: (01473) 643035) (1994).



Mike Gray joined BT Laboratories in 1972 after obtaining an honours degree in electronic engineering at Leeds University, on a scholarship. He worked initially on methods for assessing transmission performance of telephony products, including computer controlled testing. Later work included the design of colour-graphics workstations for integrated circuit design, before managing projects on computerised geographic mapping systems and public service database enquiry systems.

He has been involved in teleworking since 1988, as leader of the teleworking applications group, managing a number of major teleworking projects. Within the overall aim of developing systems to support teleworking, the projects have included thorough studies of many of the issues impacting on teleworking, the setting up of a unique experiment in homeworking for BT's operators, and helping a number of teleworking trials get started — both in BT and in other corporations. He was joint editor of 'Teleworking Explained' (published by BT/Wiley).